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## **Accurate and Efficient Salt Dome Segmentation in Seismic Imaging Based on Transformer Learning**

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### **Abstract**

This study explores transformer-based learning for accurate and efficient salt dome segmentation in seismic imaging, emphasizing transfer learning, data efficiency, and domain generalization. We adopt a Vision Transformer (ViT) architecture pretrained via masked autoencoding on over two million unlabeled seismic patches using the Seismic Foundation Model (SFM). Fine-tuning this model on as few as 1000 labeled slices yields high segmentation performance, achieving up to 95.7% accuracy and 0.91 mean Intersection over Union (mIoU) on unseen test data. Models fine-tuned on the full 4000-slice training set further improve generalization, reaching 98.3% accuracy and 0.96 mIoU. These results are validated on a curated out-of-distribution (OOD) benchmark that includes both real and synthetic seismic slices with varied geologic and amplitude characteristics. SFM-pretrained ViTs consistently outperform scratch-trained transformers and conventional convolutional neural network (CNN) baselines, even under limited supervision. Additionally, we show that applying slice-wise Z-score normalization during inference enhances model robustness under distribution shift with minimal computational overhead. Together, these findings highlight a practical and scalable approach to salt dome interpretation that combines foundation model pretraining with minimal labeling and simple adaptation strategies to achieve state-of-the-art performance, even across geologically and acoustically diverse seismic datasets.

### **Introduction**

Accurate segmentation of salt domes in seismic imaging is a fundamental task in geophysical interpretation, supporting subsurface velocity modeling, structural analysis, and hydrocarbon prospecting. Salt bodies often exhibit complex geometries, sharp discontinuities, and amplitude anomalies that vary significantly across surveys, making manual delineation both labor-intensive and prone to interpreter bias. Automated approaches—particularly convolutional neural networks (CNNs) such as U-Net ([Ronneberger et al., 2015](#))—have shown promise for pixel-wise segmentation, yet their performance often degrades when applied to data from different basins or acquisition conditions.

This degradation arises from domain shifts introduced by differences in acquisition parameters, processing workflows, and geologic context. CNNs, constrained by local receptive fields, tend to overfit

to the statistical structure of the training data, limiting their robustness in cross-survey applications (Zhang and Ma, 2019; Qiu et al., 2018).

Vision Transformers (ViTs) have emerged as a compelling alternative due to their ability to capture long-range dependencies and model global context more effectively than CNNs (Dosovitskiy et al., 2021; Vaswani et al., 2017). In geophysics, the recent development of foundation models such as the Seismic Foundation Model (SFM) has enabled large-scale self-supervised learning from millions of unlabeled seismic patches (Sheng et al., 2025). These models encode rich seismic priors that can be fine-tuned for downstream tasks—such as fault detection, facies classification, and salt interpretation—with minimal labeled data, making them well-suited for domains where annotations are costly or sparse.

However, the generalization behavior of these models under domain shift—particularly for the task of salt dome segmentation—has not been systematically evaluated. Salt structures are especially sensitive to amplitude scaling, textural differences, and noise levels, which makes this a valuable test case for studying foundation model adaptation.

In this study, we investigate the effectiveness of SFM-pretrained ViTs for salt dome segmentation under limited supervision. We compare transfer learning from SFM against several baselines, including U-Net and ViTs trained from scratch. Our results show that SFM-based models outperform traditional architectures even when fine-tuned on as few as 1,000 labeled slices, demonstrating superior data efficiency and faster convergence.

A key practical insight from our work is the role of slice-wise Z-score normalization at inference time. While normalization is a routine component of training pipelines, it is frequently overlooked during deployment in seismic interpretation workflows. Our experiments show that applying Z-score normalization at test time—without any retraining—significantly improves segmentation accuracy across seismic volumes from different domains. This finding positions normalization not merely as a preprocessing step, but as an effective, lightweight strategy for domain adaptation in seismic segmentation tasks.

## Related Work

This section reviews traditional, deep learning, and transformer-based approaches to salt dome segmentation, with a focus on model generalization and domain adaptation in seismic interpretation.

### Classical Methods for Salt Dome Segmentation

Traditional salt interpretation techniques rely on amplitude-based edge detection, texture descriptors, and semi-automated tracking of salt-sediment boundaries. Common methods include Sobel filters, gradient operators, and morphological transformations applied to seismic attributes. While effective in structured scenarios, these approaches often require interpreter supervision and are sensitive to amplitude variability and noise (Hale, 2009). Coherence and semblance attributes can highlight discontinuities but often struggle to delineate salt geometries that lack strong amplitude contrast.

### Deep Learning and U-Net-Based Approaches

Deep learning has become increasingly prominent in salt dome segmentation, driven by the success of encoder–decoder convolutional neural networks (CNNs) such as U-Net (Ronneberger et al., 2015). These models have performed well in public benchmarks like the TGS Salt Identification Challenge and have been adapted for seismic datasets (Qiu et al., 2018; Zhang and Ma, 2019). CNN-based architectures learn hierarchical spatial features from labeled data and generalize well within their training domain. However, their reliance on local receptive fields and sensitivity to training data distribution limits their robustness across surveys with different acquisition or processing conditions.

## Vision Transformers and Seismic Foundation Models

Vision Transformers (ViTs), introduced by [Dosovitskiy et al. \(2021\)](#), leverage self-attention mechanisms to model long-range dependencies and capture global contextual information—overcoming the locality constraints of CNNs. In geophysics, ViTs have been adopted through foundation models such as the Seismic Foundation Model (SFM; [Sheng et al., 2025](#)), which uses masked autoencoding ([He et al., 2022](#)) to learn rich representations from over two million unlabeled seismic patches. SFM has shown strong performance in fault detection, interpolation, facies classification, and denoising.

Additional ViT-based efforts have targeted salt dome interpretation directly. [Sansal et al. \(2025\)](#) proposed a scalable 3D ViT–MAE model trained on 63 global seismic volumes, achieving zero-shot generalization for salt delineation. Open-source initiatives such as Think [Onward \(2025\)](#) expands accessibility by releasing ViT–MAE models trained using alternative masking strategies, further enabling transformer adoption in geophysical tasks.

## Domain Generalization and Normalization

Despite architectural progress, generalization to out-of-distribution (OOD) seismic data remains a core challenge. Domain adaptation techniques—common in computer vision ([Sun and Saenko, 2016](#))—are not yet standard practice in seismic machine learning. Preprocessing steps like amplitude balancing or histogram equalization are often used inconsistently and typically applied outside of end-to-end learning pipelines.

Z-score normalization, though a standard statistical technique during model training, is rarely leveraged at inference in seismic applications. While its utility as a test-time adaptation mechanism has been explored in other domains, its potential to improve robustness in seismic segmentation remains underexamined.

Building on these observations, this work investigates how pretrained foundation models can be effectively fine-tuned for salt dome segmentation with limited supervision. We further introduce and evaluate slice-wise Z-score normalization as a lightweight, domain-agnostic test-time strategy that enhances generalization across seismic datasets with varying amplitude distributions.

## Methodology

This section outlines the experimental workflow used to evaluate transformer-based models for salt dome segmentation. We describe the dataset preparation, model architectures, training protocol, normalization strategies, and evaluation metrics. Emphasis is placed on measuring generalization across domains with varied seismic characteristics.

### Data Preparation

All experiments of training and finetuning utilize the TGS Salt Identification dataset ([Kaggle, 2018](#)), which includes 4000 grayscale seismic slices of size  $101 \times 101$  pixels, each paired with a binary mask indicating salt presence. To match the input resolution required by transformer architectures, both images and masks are up-sampled to  $224 \times 224$  using bicubic interpolation. We allocate 3500 slices for training and 500 for validation using the 80:20 split.

To isolate the impact of architecture and pretraining, no data augmentation, smoothing, or postprocessing is applied during training or inference. For out-of-distribution (OOD) evaluation, we curate a 52-slice test set drawn from a mix of real and synthetic seismic datasets. These slices span different amplitude distributions, noise levels, and geological contexts, offering a rigorous benchmark for cross-domain generalization.

### Model Architecture

We assess two primary model families: convolutional U-Nets and Vision Transformers (ViTs). The U-Net follows the canonical encoder–decoder design with five resolution levels and skip connections between matching layers. Each encoder stage consists of convolutional blocks with ReLU activation, batch

normalization, and max-pooling. The architecture begins with 64 filters and doubles the channel count at each down-sampling level.

For transformer-based models, we adopt the ViT-Base configuration (Dosovitskiy et al., 2021), comprising 12 transformer layers with 768-dimensional embeddings and 12 self-attention heads. Input images are divided into non-overlapping 16×16 patches and projected into token embeddings, followed by a custom multi-layer aggregation (MLA) decoder for pixel-wise segmentation. ViTs are trained under three regimes: (1) from scratch using all 4000 labeled slices, (2) fine-tuned from the Seismic Foundation Model (SFM), pretrained on two million seismic patches, and (3) fine-tuned using only 1000 slices, with and without normalization during inference.

All models are trained for 100 epochs using the AdamW optimizer (Loshchilov & Hutter, 2019) with an initial learning rate of  $1.5 \times 10^{-3}$  and a cosine annealing schedule. A batch size of 64 is used, and training is performed with mixed precision on four NVIDIA V100 GPUs. The training loss is defined by pixel-wise cross-entropy loss. Final model checkpoints are used for evaluation without model selection based on validation loss.

### Normalization Strategy

To address domain shifts at inference, we explore slice-wise Z-score normalization as a lightweight adaptation technique. For each 2D input slice  $X_{i,j}$ , we compute:

$$\hat{X}_i = \frac{X_{i,j} - \mu_X}{\sigma_X}$$

where  $X_{i,j}$  denotes the pixel intensity at location (i,j) and  $\mu_X$ ,  $\sigma_X$  are the mean and standard deviation of the slice. This normalization is applied only at inference unless otherwise noted.

Although Z-score normalization is widely used during training, its role at inference is often underexplored in seismic applications. In our experiments, we include a ViT model fine-tuned on 1000 labeled slices, with slice-wise normalization applied during both fine-tuning and inference. While this model performs comparably to its non-normalized counterpart on in-distribution validation data, it demonstrates notably improved robustness under domain shift—highlighting normalization's value as a lightweight yet effective test-time adaptation strategy.

### Evaluation Metrics

We evaluate segmentation performance using three criteria. Pixel Accuracy measures the proportion of correctly labeled pixels across the image. Mean Intersection over Union (mIoU) computes the overlap between predicted and ground-truth regions averaged across salt and background classes. Finally, qualitative inspection is conducted to visually assess structural accuracy and salt boundary delineation.

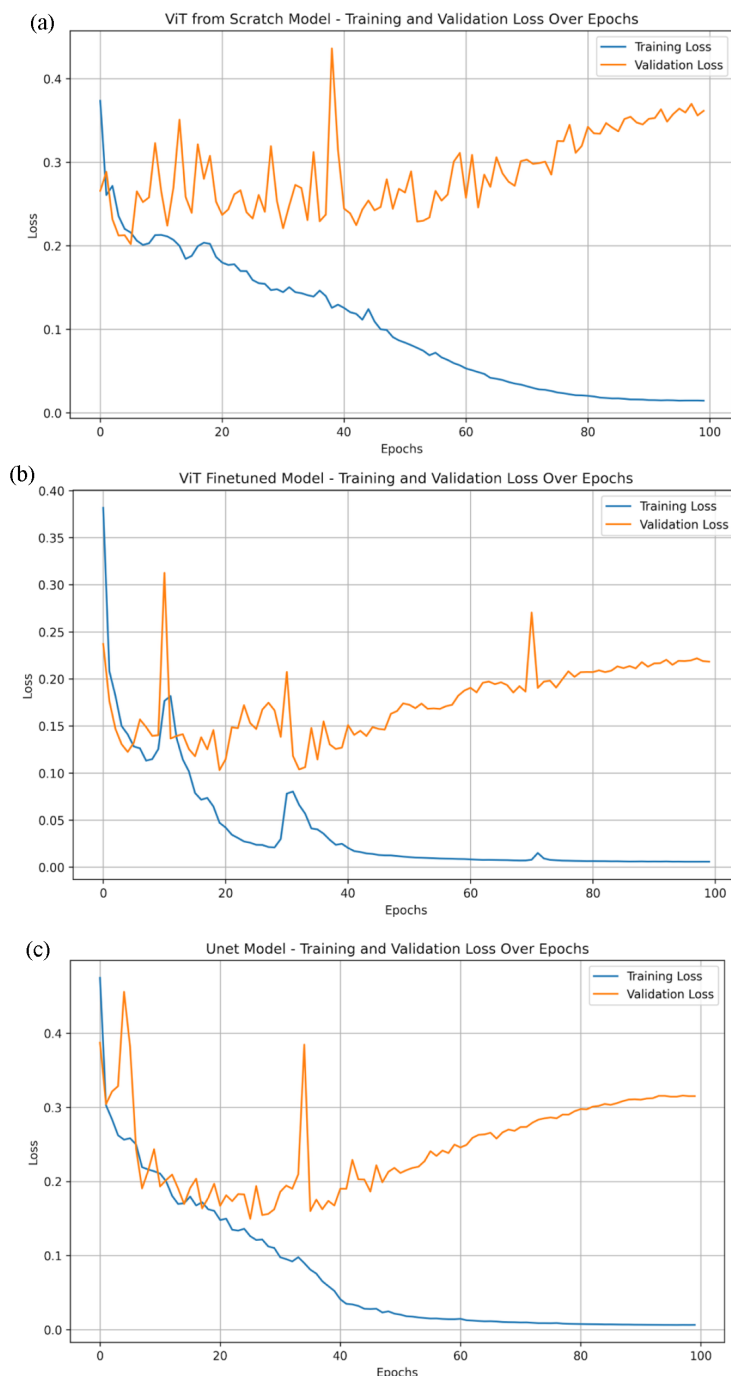
All metrics are reported on both the validation set and the external OOD test set. Each model is trained and evaluated once; no ensembling or test-time augmentation is applied.

## Results

This section presents a comprehensive evaluation of Vision Transformer (ViT) models for salt dome segmentation, focusing on training dynamics, validation accuracy, and generalization to out-of-distribution (OOD) seismic data. We compare models trained from scratch, models fine-tuned on varying amounts of labeled data, and the impact of inference-time normalization. The analysis is structured to highlight both quantitative metrics and qualitative behavior, providing insight into model performance across data regimes and geological scenarios. Together, these results demonstrate the benefits of foundation model pretraining and lightweight adaptation techniques in achieving scalable, high-fidelity salt interpretation.

## Training Dynamics and Data Efficiency

To evaluate learning behavior and architectural capacity, we examine the training and validation loss curves for ViT and U-Net models [Fig. 1](#). The ViT model trained from scratch shows a steep decline in training loss but fails to significantly reduce validation loss, indicating classic overfitting behavior due to the lack of pretrained feature priors. In contrast, the ViT model fine-tuned on 4000 labeled slices demonstrates stable convergence, with low final losses on both training and validation sets—highlighting the benefit of initializing from a pretrained seismic representation.



**Figure 1—Training and validation loss curves for: (a) ViT from scratch, (b) ViT fine-tuned on 4000 slices, and (c) U-Net. These plots illustrate convergence dynamics and the impact of model capacity and pretraining.**

The U-Net model, serving as a classical convolutional baseline, shows early plateauing in both training and validation loss curves. This saturation suggests limited representational capacity for modeling complex salt geometries, particularly when compared to transformer-based architectures.

These trends underscore the value of transformer models—especially when pretrained—both in terms of convergence stability and learning capacity. While more detailed comparisons of generalization and performance under data constraints are presented in subsequent sections, the training behavior alone reflects a clear advantage for pretrained ViTs.

### Validation Performance

We assess all models on their respective validation sets to evaluate segmentation performance under in-distribution conditions. For the ViT models trained on the full TGS dataset (4000 slices), validation is performed on a held-out set of 500 slices. For the ViT model fine-tuned on only 1000 slices, 200 slices are used for validation, reflecting an 80:20 split. This setup enables us to evaluate both absolute performance and data efficiency.

Table 1 summarizes the accuracy and mean Intersection over Union (mIoU) across models. The ViT model fine-tuned on all 4000 slices achieves 96.12% accuracy and 0.8979 mIoU, setting the performance upper bound. Interestingly, the ViT trained on only 1000 labeled slices (with 200 used for validation) achieves 97.15% accuracy and 0.9286 mIoU—slightly exceeding the full-data model. This result, while promising, may reflect slight overfitting to the validation set due to limited diversity.

**Table 1—Validation set accuracy and mean IoU across all models.**

| Model                      | Accuracy (%) | Mean IoU |
|----------------------------|--------------|----------|
| U-Net                      | 93.85        | 0.8472   |
| ViT (scratch)              | 91.35        | 0.7890   |
| ViT (1000 slices, 200 val) | 97.15        | 0.9286   |
| ViT (1000 + Z-score)       | 97.21        | 0.9303   |
| ViT (4000 slices)          | 96.12        | 0.8979   |

When slice-wise Z-score normalization is applied at inference, the 1000-slice model achieves an even higher mIoU of 0.9303, suggesting enhanced robustness without requiring additional training. By contrast, ViT trained from scratch shows reduced performance (91.35% accuracy, 0.7890 mIoU), confirming poor generalization in the absence of pretraining. U-Net, while outperforming the scratch ViT, still lags behind all pretrained ViT variants.

To visualize model performance, Fig. 2 shows a representative validation slice from the TGS dataset. The ViT (1000 slices) prediction closely matches the salt geometry in the ground truth, illustrating the model's ability to accurately recover structural boundaries even with limited supervision.



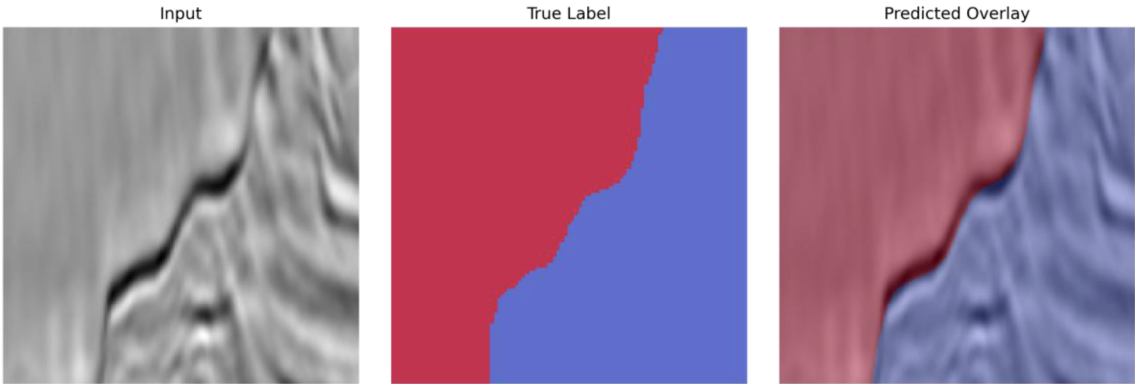


Figure 2—Validation slice prediction using ViT fine-tuned on 1000 slices. The model output (center) closely matches the input slice (left) and ground-truth mask (right), accurately delineating the salt body.

Generalization to Unseen Data

To evaluate model robustness under domain shift, we test all Vision Transformer (ViT) variants on a curated out-of-distribution (OOD) benchmark comprising 52 seismic slices. These slices include both synthetic and real examples from geologically distinct domains and acquisition conditions, posing challenges not seen during training. Our analysis includes ViT models fine-tuned on the full 4000-slice training set using pretrained seismic features, ViT models fine-tuned on 1000 labeled slices, and ViT models trained from scratch on 4000 slices without pretraining. Each model is evaluated both with and without inference-time slice-wise Z-score normalization.

**Quantitative Results.** Table 2 summarizes the accuracy and mean intersection over union (mIoU) metrics for all models on the unseen test set. The ViT model trained from scratch performs poorly without normalization, achieving only 47.57% accuracy and 0.2886 mIoU. However, applying Z-score normalization at inference yields a dramatic improvement to 93.47% accuracy and 0.8592 mIoU, underscoring the importance of amplitude standardization in the absence of learned priors.

Table 2—Generalization results on the unseen test set (52 slices). Accuracy and mIoU are reported with and without slice-wise Z-score normalization applied at inference.

| Model                         | Accuracy (%) | Mean IoU |
|-------------------------------|--------------|----------|
| ViT (Scratch, no Z-score)     | 47.57        | 0.2886   |
| ViT (Scratch, +Z-score)       | 93.47        | 0.8592   |
| ViT (1000 slices, no Z-score) | 57.56        | 0.2878   |
| ViT (1000 slices, +Z-score)   | 95.73        | 0.9113   |
| ViT (4000 slices, no Z-score) | 57.57        | 0.2878   |
| ViT (4000 slices, +Z-score)   | 98.35        | 0.9581   |

The ViT model fine-tuned on 1000 slices similarly benefits from normalization. Without normalization, it achieves 57.56% accuracy and 0.2878 mIoU, while normalization boosts performance to 95.73% accuracy and 0.9113 mIoU, approaching the performance of full-data fine-tuning.

Finally, the ViT model fine-tuned on 4000 labeled slices achieves the highest generalization performance overall, with 98.35% accuracy and 0.9581 mIoU when normalization is applied. Notably, its performance drops to 57.57% accuracy and 0.2878 mIoU without normalization, despite having access to the full labeled dataset.

These results suggest that even well-supervised models benefit from test-time normalization, and that pretrained models paired with normalization can match or exceed the performance of larger supervised setups under distribution shift.

**Qualitative Comparison.** To complement the quantitative evaluation, we present qualitative segmentation results on two representative unseen slices—Test Slice 1 and Test Slice 2—illustrating model behavior under domain shift. These slices exhibit distinct geological patterns and serve as case studies for boundary resolution and structural accuracy.

Test Slice 1 (Fig. 3) corresponds to a moderately complex salt structure with a sharp upper interface and a diffuse base. The ViT model trained from scratch fails to capture the salt geometry, producing fragmented predictions and missing the upper boundary entirely. The ViT model fine-tuned on 4000 labeled slices recovers both salt interfaces with high fidelity, capturing the full structure with minimal artifacts. The ViT model fine-tuned on 1000 slices shows improved edge continuity over the scratch model but remains noisy in deeper zones, particularly near the base salt boundary.

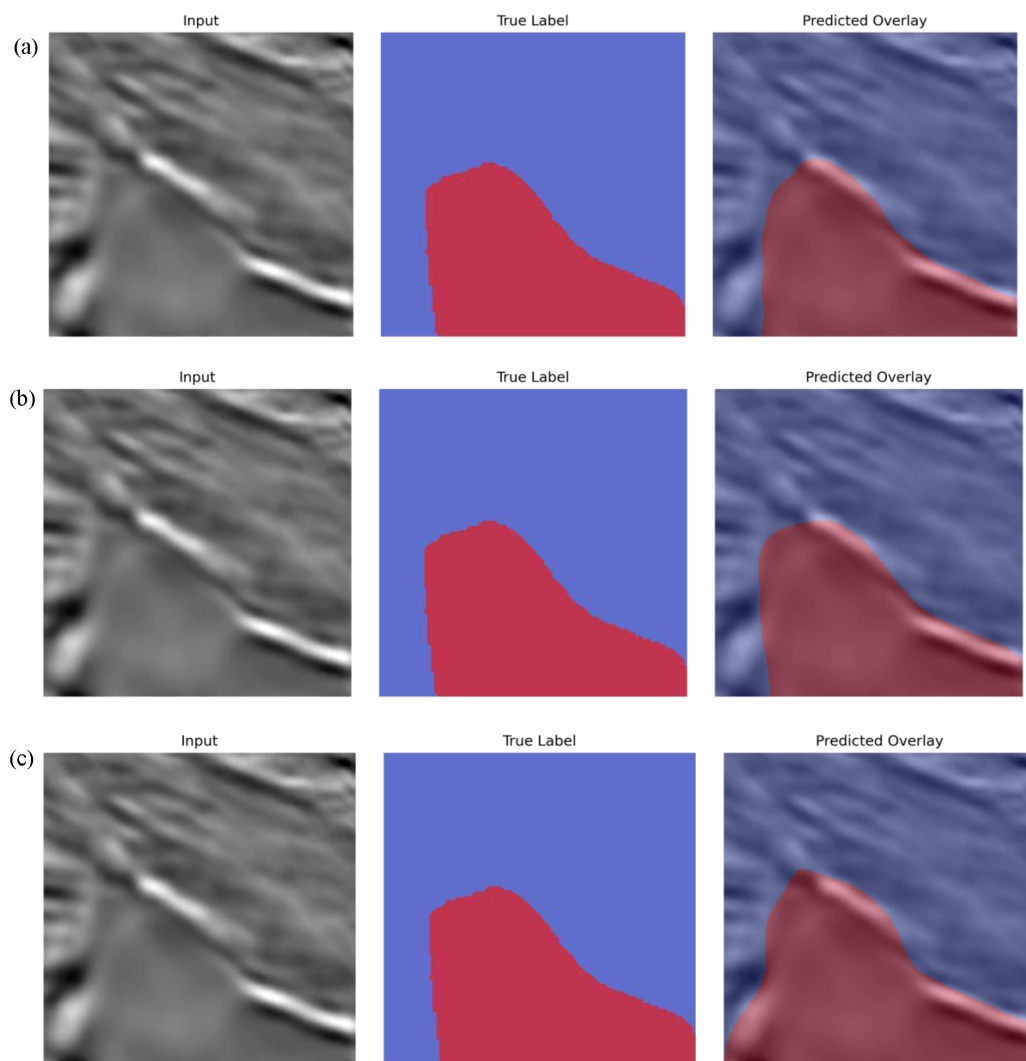
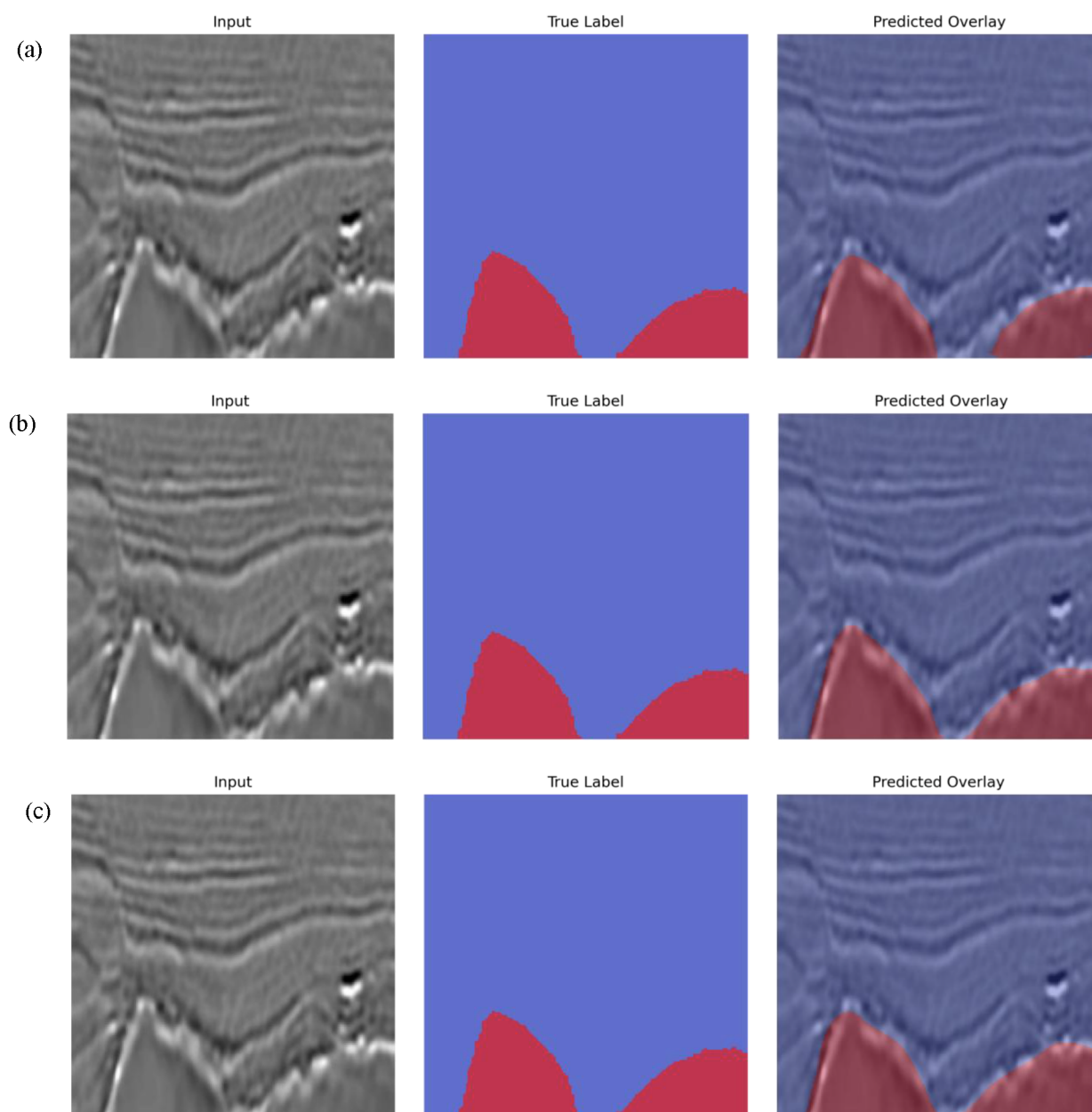


Figure 3—Segmentation results on Test Slice 1. From left to right: Input and Ground Truth. From top to bottom: (a) ViT (Scratch), (b) ViT (4000 slices), (c) ViT (1000 slices).

Test Slice 2 (Fig. 4) presents a more challenging scenario involving dual salt bodies and lateral amplitude variation. The scratch-trained model undersegments the left salt body and exhibits spurious leakage into adjacent non-salt regions. The 1000-slice model performs better, capturing both structures, but introduces



blurring along the salt–non-salt interface. In contrast, the 4000-slice fine-tuned model reconstructs both bodies with precise geometry, clean separation, and consistent flank delineation—demonstrating its superior generalization capacity in complex domains.



**Figure 4—Segmentation results on Test Slice 2. From left to right: Input and Ground Truth. From top to bottom: (a) ViT (Scratch), (b) ViT (4000 slices), (c) ViT (1000 slices).**

These visual comparisons reinforce the findings in [Table 2](#), confirming that transformer-based models benefit significantly from pretraining and larger-scale supervision. Notably, even models trained on limited labeled data can produce accurate salt dome segmentation when guided by pretrained seismic representations.

## Discussion

This study demonstrates the effectiveness of leveraging a pretrained Seismic Foundation Model (SFM) for salt dome segmentation through minimal supervised fine-tuning. The findings support three central insights:

the value of foundation model pretraining for geologic feature recognition, the benefits of inference-time normalization for domain generalization, and the viability of data-efficient fine-tuning for high-accuracy seismic interpretation.

First, ViT-based architectures initialized from SFM exhibit strong generalization to out-of-distribution (OOD) data, even when fine-tuned on as few as 1000 labeled slices. These models achieve high segmentation accuracy on both validation and unseen test sets, recovering salt boundaries with notable structural fidelity. This is particularly evident in complex cases such as Test Slice 2 (Fig. 4), where the fine-tuned models successfully delineate dual salt bodies and preserve lateral continuity. In contrast, the ViT trained from scratch fails to resolve even the primary geometry—highlighting the importance of pretraining on diverse seismic structures.

Second, inference-time slice-wise Z-score normalization emerges as a lightweight yet impactful strategy for improving robustness under amplitude variability. Applied uniformly across all model variants, normalization consistently boosts generalization performance—elevating scratch-trained ViT accuracy from 47.57% to 93.47%, and fine-tuned models to near-perfect segmentations (Table 2). This confirms that amplitude standardization remains a critical, often overlooked component of seismic deployment workflows. Notably, even the ViT fine-tuned on 4000 slices experiences a drastic drop in performance without normalization, suggesting that large-scale supervision alone is insufficient to handle unseen distributions.

Third, the qualitative results align closely with the quantitative metrics and offer practical interpretability. In Test Slice 1 (Fig. 3), the scratch-trained model misclassifies base salt zones and introduces fragmentation, while the 1000-slice model reduces noise but remains inconsistent in deeper sections. Only the 4000-slice fine-tuned ViT cleanly resolves both upper and lower boundaries with minimal artifacts. These visual patterns reaffirm that foundation model priors enable not only sharper salt delineation but also better suppression of spurious features—critical in production environments where visual confidence is paramount.

Despite these strengths, several challenges remain. Current models operate on individual 2D slices, potentially missing volumetric context and inter-slice continuity that 3D models could capture—especially important for delineating complex salt geometries. While inference is efficient and scalable, the interpretability of transformer attention mechanisms remains limited, warranting further study through techniques such as attention maps or saliency visualizations—especially in high-stakes exploration scenarios. Additionally, while Z-score normalization proves effective, future work should investigate adaptive normalization or test-time domain adaptation strategies that dynamically respond to geological and amplitude variability.

Overall, this study makes a compelling case for combining pretrained seismic transformers with lightweight normalization strategies to deliver accurate, efficient, and generalizable salt dome segmentation. The proposed approach balances geologic fidelity with operational practicality, paving the way for scalable, automated interpretation across diverse seismic settings.

## Conclusion

This study introduces a robust and scalable framework for salt dome segmentation by combining pretrained Vision Transformers (ViTs) with inference-time slice-wise Z-score normalization. By leveraging the Seismic Foundation Model (SFM), our approach achieves robust segmentation performance with minimal labeled data—maintaining structural fidelity even under challenging domain shifts. These models not only outperform scratch-trained ViTs but also exceed the performance of classical CNN-based architectures like U-Net—achieving strong generalization with minimal supervision.

A central finding is the critical role of inference-time normalization in enhancing model robustness. Z-score normalization offers a lightweight yet powerful enhancement, enabling even underperforming

models to generalize effectively—without additional training or hyperparameter tuning. This underscores its practical importance as a lightweight test-time adaptation strategy for real-world seismic interpretation.

Together, these results demonstrate that pretrained seismic transformers, when paired with simple yet effective normalization techniques, enable accurate, efficient, and domain-generalizable salt dome segmentation. The proposed approach offers a practical path forward for scaling automated interpretation workflows across diverse geologic settings, bridging the gap between data-driven modeling and operational geoscience.

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